



Echinococcus granulosus and Hymenolepis nana — MCQ Practice

Microbiology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A-E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. What is the common name of Echinococcus granulosus?

EASY

- A. Beef tapeworm
- B. Pork tapeworm
- C. Dog tapeworm
- D. Fish tapeworm
- E. Dwarf tapeworm

2. In which organ does the adult Echinococcus granulosus worm live in its definitive host?

EASY

- A. Liver of humans
- B. Lungs of herbivores
- C. Colon of cats
- D. Muscle of sheep
- E. Jejunum and duodenum of dogs/canids

3. What is the larval stage of Echinococcus granulosus called?

EASY

- A. Cysticercus
- B. Metacercaria
- C. Cysticercoid
- D. Oncosphere
- E. Hydatid cyst

4. Which drug is the treatment of choice for Hymenolepis nana?

EASY

- A. Metronidazole
- B. Pyrantel pamoate
- C. Doxycycline
- D. Praziquantel
- E. Ivermectin

5. What is "hydatid sand"?

MEDIUM

- A. Calcified debris in a chronic cyst
- B. Sand-like material found in endemic soil
- C. Free protoscolices lying at the bottom of the hydatid cyst
- D. Eggs of E. granulosus found in sheep feces
- E. A diagnostic stain artifact

6. Approximately what percentage of hydatid cysts develop in the liver?

MEDIUM

- A. 5–10%
- B. 60–75%
- C. 20–25%
- D. 90–95%
- E. 40–50%

7. Why does the lung act as a "second filter" for Echinococcus granulosus embryos?

MEDIUM

- A. The lung has the highest blood flow of any organ
- B. Embryos that pass through the liver enter the right heart and lodge in pulmonary capillaries
- C. Embryos preferentially target respiratory epithelium
- D. The lung is the first organ reached after ingestion
- E. Lymphatic drainage carries embryos directly to the lung first

8. What is the recommended early-stage treatment for an uncomplicated hydatid cyst?

MEDIUM

- A. Immediate open surgical excision only
- B. Observation alone with no intervention
- C. Praziquantel monotherapy
- D. PAIR (puncture, aspiration, injection, reaspiration)
- E. Radiotherapy

9. Why is pre- and post-operative albendazole recommended with surgery for hydatid disease?

MEDIUM

- A. To treat concurrent bacterial infection
- B. To reduce recurrence and treat any spilled protoscolices around the surgical site
- C. To prevent anaphylaxis directly
- D. To shrink the cyst before any surgery is needed
- E. To replace surgery entirely in all cases

10. A patient with a ruptured hydatid cyst develops sudden hypotension, wheeze, and urticaria. What is the mechanism?

MEDIUM

- A. Direct bacterial toxin release
- B. Acute pulmonary embolism from cyst debris
- C. Anaphylactic reaction to parasite antigens released from the cyst fluid
- D. Vasovagal response to pain
- E. Hemorrhagic shock from cyst bleeding

11. Which laboratory finding may be seen with cystic echinococcosis on a blood count?

MEDIUM

- A. Marked neutropenia
- B. Pancytopenia
- C. Eosinophilia of 20–25%
- D. Lymphocytosis >90%
- E. Basophilia >50%

12. What technique is used to detect excreted protoscolices in urine or sputum?

MEDIUM

- A. Lactophenol cotton blue (LPCB) staining
- B. Gram stain
- C. Ziehl-Neelsen stain
- D. Giemsa stain only
- E. India ink preparation

13. Why is Hymenolepis nana infection most common in school children and institutional populations?

HARD

- A. Because the entire life cycle completes in one host, allowing easy person-to-person/fecal-oral spread in crowded settings
- B. Because it requires an insect vector found mainly indoors
- C. Because children uniquely lack the receptor for infection
- D. Because adults are naturally immune after a single exposure
- E. Because only undercooked food served in institutions carries the parasite

14. How many hooklets rows does the Hymenolepis nana scolex rostellum bear?

MEDIUM

- A. A single row
- B. Two rows
- C. Four rows
- D. No hooklets, only suckers
- E. Three concentric rows

15. What happens to Hymenolepis nana eggs once passed in feces?

MEDIUM

- A. They must embryonate in soil for 30 days before becoming infective
- B. They require a beetle intermediate host to mature
- C. They can survive for years in soil, unlike other cestode eggs
- D. They are immediately infective but survive less than 10 days in the external environment
- E. They require desiccation before becoming infective

16. A 35-year-old immigrant from a sheep-farming region presents with a solitary 8 cm liver cyst with daughter cysts on ultrasound; the cyst is deemed too complex for PAIR. What is the most appropriate definitive management?

HARD

- A. Surgical excision with care to avoid rupture, plus perioperative albendazole
- B. PAIR alone regardless of complexity
- C. Albendazole monotherapy for 6 months with no procedure
- D. Immediate percutaneous drainage without antihelminthic cover
- E. Observation only, since most cysts resolve spontaneously

17. Which feature distinguishes amebic liver abscess management considerations from hydatid liver cyst management, specifically regarding aspiration?

HARD

- A. Hydatid cysts can be freely aspirated with no special precautions
- B. Amebic abscesses cannot be aspirated under any circumstances
- C. Both carry identical aspiration risk profiles
- D. Hydatid cysts never require any imaging before intervention
- E. Hydatid cyst aspiration/rupture risks anaphylaxis and secondary cyst seeding, requiring extreme caution not required for simple drainage of other liver lesions

18. Why can pathogenic Echinococcus granulosus cysts cause symptoms resembling a slow-growing tumor regardless of organ involved?

HARD

- A. The parasite actively invades and destroys surrounding tissue like a malignancy
- B. The cyst secretes growth factors that stimulate tumor formation
- C. Cysts always calcify rapidly, mimicking bone tumors
- D. Clinical disease results mainly from pressure effects of the gradually enlarging, fibrous-encapsulated cyst
- E. The host mounts no immune response to the cyst at all

19. A patient with cerebral hydatid disease develops new-onset seizures. What is the underlying mechanism?

HARD

- A. Direct neurotoxin secreted by the scolex
- B. Raised intracranial pressure and local mass effect from the enlarging cyst
- C. Autoimmune encephalitis triggered by cyst antigens
- D. Hemorrhagic stroke from cyst rupture into a vessel
- E. Meningitis caused by bacterial superinfection of the cyst

20. Why is molecular (PCR) diagnosis particularly useful for Hymenolepis nana or Echinococcus granulosus compared to morphology alone in some contexts?

HARD

- A. Morphologic diagnosis is impossible for all cestodes without PCR
- B. PCR is the only method that can be performed on fresh stool
- C. PCR replaces the need for any imaging in all cases
- D. PCR is required because these organisms have no identifiable eggs
- E. It increases sensitivity/specificity, especially where egg/cyst morphology may be equivocal or mixed infections are suspected

✓ Answer key

Q	Answer	Difficulty	Why
1	C	EASY	<i>E. granulosus</i> is known as the dog tapeworm, with canids as the definitive host.
2	E	EASY	The adult worm lives in the small intestine (jejunum/duodenum) of dogs and other canids.
3	E	EASY	The larval form that develops in intermediate hosts (including humans) is the hydatid cyst.
4	D	EASY	Praziquantel is the drug of choice, acting on both adults and cysticercoids.
5	C	MEDIUM	Hydatid sand refers to individual free scolices that settle at the bottom of the large hydatid cyst.
6	B	MEDIUM	The liver, as the first filter for the embryo, hosts about 60-75% of hydatid cysts.
7	B	MEDIUM	Embryos passing the hepatic filter travel via the right heart to the pulmonary capillaries, making the lung the second filter.
8	D	MEDIUM	PAIR is the treatment of choice in early stages before cysts become complicated.
9	B	MEDIUM	Chemotherapy with albendazole around curative surgery helps reduce recurrence, which is otherwise common.
10	C	MEDIUM	Cyst fluid antigens can sensitize the host, and rupture can precipitate life-threatening anaphylaxis.
11	C	MEDIUM	Generalized eosinophilia of 20-25% may be found in cystic echinococcosis.
12	A	MEDIUM	LPCB staining is used to demonstrate scolices excreted into sputum or urine.
13	A	HARD	With no intermediate host required and eggs immediately infective, crowded settings like schools/institutions favor fecal-oral spread.
14	A	MEDIUM	The retractile rostellum bears a single row of hooklets, in addition to 4 suckers.
15	D	MEDIUM	<i>H. nana</i> eggs are immediately infective upon passage and survive less than 10 days outside the host, consistent with direct person-to-person spread.
16	A	HARD	Complicated cysts are managed surgically, with meticulous technique to avoid rupture, combined with perioperative albendazole to reduce recurrence.
17	E	HARD	Unlike many other liver lesions, hydatid cyst fluid contains antigenic material and viable protoscolices, so aspiration/rupture carries unique risks of anaphylaxis and dissemination — a key safety teaching point.
18	D	HARD	Because the cyst grows slowly within a fibrous capsule, it causes symptoms through mass effect and pressure rather than invasive destruction — mimicking a slow-growing tumor.
19	B	HARD	Cerebral hydatid cysts produce raised intracranial pressure and local mass effect, which can manifest as seizures.
20	E	HARD	While both organisms have characteristic morphology (hydatid cyst imaging appearance, <i>H. nana</i> 's distinctive egg), PCR adds sensitivity/specificity particularly in ambiguous or mixed-infection scenarios.